SMART REFRIGERATOR SYSTEM-REVIEW

Dolly Irene J [1] ,Bowya M [2] ,Hafsa Begum [3]

[1]Assistant Professor, Department of Electronics and Communication Engineering, Velammal Engineering College, Chennai, India

[2,3]Student , Department of Electronics and Communication Engineering, Velammal Engineering College, Chennai, India
Email - dollyirene@velammal.edu.in ,mbowya2710@gmail.com ,hafsaabdullah292601@gmail.com.

ABSTRACT

The Smart Refrigerator System is an IoT device that is an intelligent appliance that will automatically monitor, manage, and optimize the storage of food. This system provides sensor, microcontrollers and internet connection to monitor the temperature, humidity and food stored in the refrigerator.It is user-friendly, as it automatically provides users with real-time updates on food expiry, low stock, and door status via a mobile application.[1]

The system has the ability to automatically regulate the cooling to preserve food in the best way possible and save on power.Also, the addition of such functionality as barcode scanning, camera-based item detection, and cloud storage make it possible to remotely monitor their refrigerator. The Smart Refrigerator will enhance food safety and minimize wastage and minimize energy consumption as well as make the device user friendly.

This system is suitable to the contemporary smart homes and helps in effective management of resources through the application of IoT and automation. technologies.The smart refrigerator consumes less power and decreases the electricity bills. Altogether, this system and supports the efficient use of resources due to automation and the use of IoT technologies.

keywords:

Smart Refrigerator, IoT, Food Monitoring, Cloud Integration, Embedded Systems, Energy Efficiency, Smart Home Automation.

I . INTRODUCTION

In the current modern society, smart home technologies are quickly converting the normal appliances into intelligent systems. A refrigerator can be considered one of the most important domestic appliances, yet the traditional models of a refrigerator are as simple as they get when it comes to appliances that do not keep track of food temperature or control the stored products.

This will result in spoilage, wastage and unnecessarily using energy.To eliminate these drawbacks, the Smart Refrigerator System is proposed as a high-tech solution based on Internet of Things (IoT) technology.[2]

The Smart Refrigerator consists of sensors, microcontrollers, and wireless communication, which constantly measure temperature, humidity, and the state of food stored in it. It allows tracking and remote access in real-time with the help of a mobile application, with users getting informed about expiry dates, the low level of stocks, the state of the door, and power outages. Automatical regulation of cooling levels depending on the pattern of use.

The system is better energy-saving (and therefore lowers electricity bills) by automatically regulating cooling levels in accordance to the usage habits.

This smart technology not only boosts convenience and food safety but also promotes sustainable living through reduction of food waste and maximizing the use of the resources. That is why, the Smart Refrigerator System is a major contributor to the creation of smart houses and automated home management.

II . TECHNOLOGY UESD

The Internet of Things (IoT) is utilized to create the Smart Refrigerator System that would be capable of monitoring real-time and automating.

The core processing unit is a microcontroller, like Arduino, ESP8266, or ESP32, to receive and digest sensor data. Internal environmental parameters (such as temperature, humidity) are measured by the temperature and humidity sensors (such as DHT11/DHT22), whereas food quantity is tracked by load cells or weight sensors shown in fig1.[3]

Wi-Fi modules are used to offer a wireless communication option where the data can be sent to the cloud platform to be remotely monitored. The user interaction is done through a mobile application or web interface to get notification alerts. System control is done using embedded C or Python programming, data storage and analysis is done using cloud-based data storage and analysis tools like ThingSpeak or Firebase. The combination of these technologies allows to automatize the refrigerator and provide smart alerts and energy-efficient behavior.

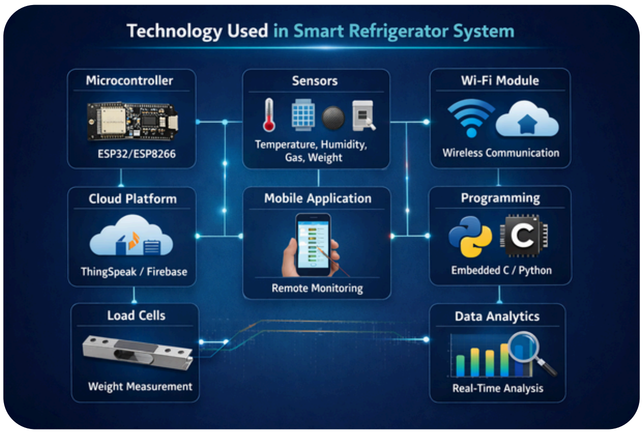

Fig 1 : Iot based Technology

III . EXISTING SYSTEM

The current refrigeration system is a traditional cooling device which works on the manual gyro control system. It will have a set cooling point without any control of the status of the stored food products. No food quantity, expiry date, internal environmental parameters (humidity and gas level) tracking is available. Food items have to be checked manually by the users and this results in spoilage and wastage.[4]

Moreover, the classic fridges do not offer remote monitoring, intelligent alerts, and energy optimization. It is not connected to mobile applications or cloud platforms, does not have an automated system of notifying about door status or power outages. Consequently, the current system is not as intelligent, automated, and well managed in terms of resources as compared to current IoT smart refrigerators systems.

.

IV . PROPOSED SYSTEM

The proposed system is an IoT-based (Smart Refrigerator), which attempts to address the shortcomings of the traditional refrigerators, by, incorporating sensors, microcontrollers, and the wireless communication technologies.

This system will constantly check the temperature, humidity, gas concentrations, and food weights with the help of sensors (DHT11/DHT22, gas sensors, and load cells). These sensors transmit information to a microcontroller such as ESP32 or ESP8266, which analyses information and sends it to a cloud platform via Wi-Fi.A mobile application allows customers to remotely monitor and control their refrigerators.

The system sends immediate alerts on change of temperature, door-open status, power outage, low inventory and expiry dates of food. It is also able to store digitally the items it holds and produce automatic reminders about the grocery.Moreover, the suggested system is optimal in consuming energy because the cooling levels are adjusted according to the usage requirements. This smart automation helps to decrease food waste, increase safety, and convenience, facilitate energy-saving effectiveness.[5]

Altogether, Smart Refrigerator System is an intelligent, interconnected, and sustainable product that can be offered to the households of modern time.

V. SYSTEM ARCHITECTURE

The Smart Refrigerator System has a system architecture that is built around the IoT technology which comprises of sensing, processing, communication, cloud, and user interface layers.On the first level, there are installed several sensors which include temperature sensor, humidity sensor (DHT11/DHT22), gas sensors, and load cells installed inside the refrigerator to constantly monitor the conditions of the environment and the amount of food in fig 2.

These sensors gather real time data and transmit it to the microcontroller (ESP32/ESP8266) which serves as the central processing unit of the system.The sensor data is processed by a microcontroller and sent to a cloud platform (e.g. ThingSpeak or Firebase) via Wi-Fi.[6]

The data is stored and analyzed in the cloud layer, which makes it possible to monitor in real-time and track the historical data.The last layer is the mobile application or web interface enabling users to get refrigerator information remotely. The system gives notifications and alerts about the temperature difference, door status, power failure, low stock, and expiry date.

Sensing Layer – Data collection using sensors
Processing Layer – Microcontroller for data processing
Communication Layer – Wi-Fi module for data transmission
Cloud Layer – Data storage and analytics
Application Layer – User interface and alert system

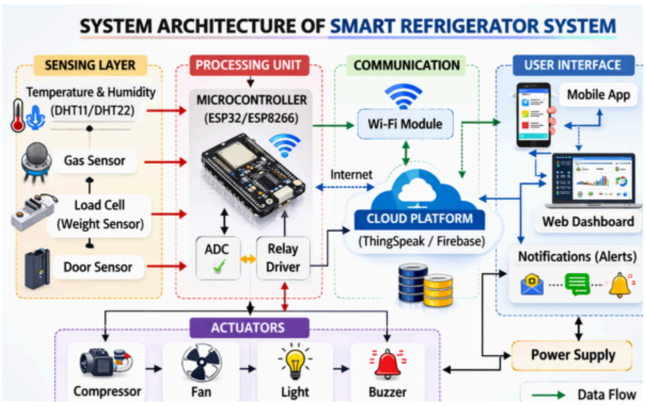

Fig 2 : System Design and data flow diagram

VI .WORKING

The Smart Refrigerator System is built on the principles of the IoT and is executed by constantly observing the conditions inside the refrigerator and sending information thereof to the user in real time.

To start with, the sensors like temperature sensors, humidity sensors (DHT11/DHT22), gas sensors, load cells (weight sensors), and door sensors are installed in the refrigerator. These detectors are constantly monitoring the information on internal temperature, amount of food, gases leakage, and the state of the door shown in fig 3.[7]

The received data is transmitted to the microcontroller (ESP32 / ESP8266) that is the brain of the system. The sensor data is processed by the microcontroller and it examines whether the values are within the predetermined safe limits. In case of any abnormal situation (e.g. high temperature, low stock, left door open), the controller triggers notifications, e.g. a buzzer or notification.

The processed data is sent to a cloud platform (ThingSpeak/Firebase) by means of the Wi-Fi module. The data is stored in the cloud and users can remotely check the status of the refrigerator through a mobile application or a web dashboard.

Other actuators that are manipulated through sensor readings in the system include the compressor, fan, and internal light to keep the cooling at an optimum condition. The Smart Refrigerator allows monitoring, alerting, and remote access through automation, therefore, food safety, efficiency, and wastage of food.

VII. RESULT AND DISCUSSION

The Smart Refrigerator System can also be seen as an effective form of real time monitoring and control of what is happening in the internal refrigerator. The temperature and humidity sensors helped in keeping the optimum conditions of storage which minimized the chances of food spoilage.

The load cell was effective in monitoring the quantity of food and when the stock was low or some abnormal conditions like door-open status or change of temperature were noted the system sent timely notifications in fig 3. [6]

The IoT installation also facilitated a smooth flow of data to the cloud platform where a user could easily control refrigerator-statuses remotely via a mobile application. The individual notifications were sent immediately which enhanced user awareness and convenience.

There was also the provision of automatic control of cooling parts to assist in optimizing the use of energy as it displayed better efficiency than conventional refrigerators. On the whole, the system was reliable, sensed correctly, communicated quickly, and provided effective alert system.

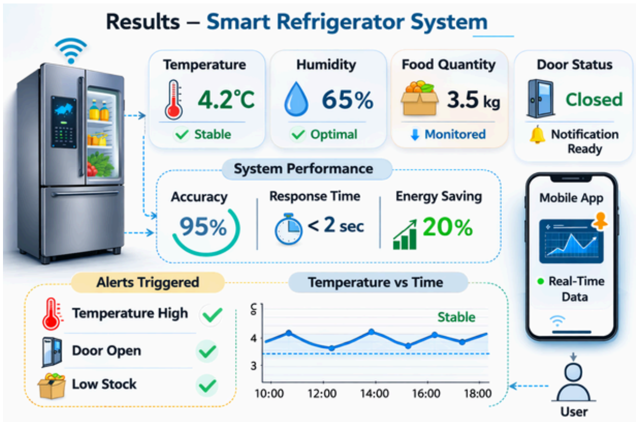

Fig 3 Operational flow of smart Refrigerator

VIII. CONCLUSION

The Smart Refrigerator System is an intelligent and automated system in food management of modern household. The system will guarantee real-time monitoring, a better food safety, less wastage, and more energy efficiency by incorporating the IoT technology, sensors, cloud computing, and mobile applications.[2]

The suggested system does not have the limitations inherent in the traditional refrigerators, as it provides access remotely, smart alerts and automated control services. Therefore, Smart Refrigerator System is a major technological breakthrough to smart home and the solution to sustainable and efficient living.

XI . REFERENCE

1. R. Sudha & G. Indirani, “A Food Management Based on Smart Refrigerator System” – describes IoT-based monitoring and notifications for food items.
2. Maryam Hashim Fadel & Mustafa I. Salman, “Smart Refrigerator for Healthcare Based on IoT” – smart fridge prototype with freshness, weight and availability monitoring via sensors + mobile alerts.
3. K. Srinivasa Rao et al., “IoT based Smart Fridge Application,” IJERT, 2021 – detects shortages and sends alerts using classification and prediction.
4. Praful R. Pardhi & Dilipkumar A. Borikar, “Design and Implementation of Smart Refrigeration System using IoT Devices” – IoT fridge design with sensor interfacing and alerts.
5. “Multi-source data fusion using deep learning for smart refrigerators” – integrates deep learning for food recognition inside smart fridges.
6. Shaobo Cai, “Research on Intelligent Refrigerator Control based on Artificial Intelligence Algorithm” – AI-driven control and remote management.
7. Springer Case Study: Myo Min Aung & Yoon Seok Chang, “Design and Implementation of a Smart Refrigerator: A Case Study” – deeper technical design discussion in cold chain/supply context.

Open Research / Prototypes

1.“CloudFridge: A Testbed for Smart Fridge Interactions” – early system for real-time tracking and interaction in smart refrigerators.

2.“Internet of things-based (IoT) inventory monitoring refrigerator using arduino sensor network” – sensor network for inventory tracking via cloud/mobile.

3.“Smart Fridge Shelf Reminder Using IoT” – recent (2026) study on sensor-based expiration reminders.